

GENIFY

LECTURE 13

13-August-2025

TOPICS COVERED:

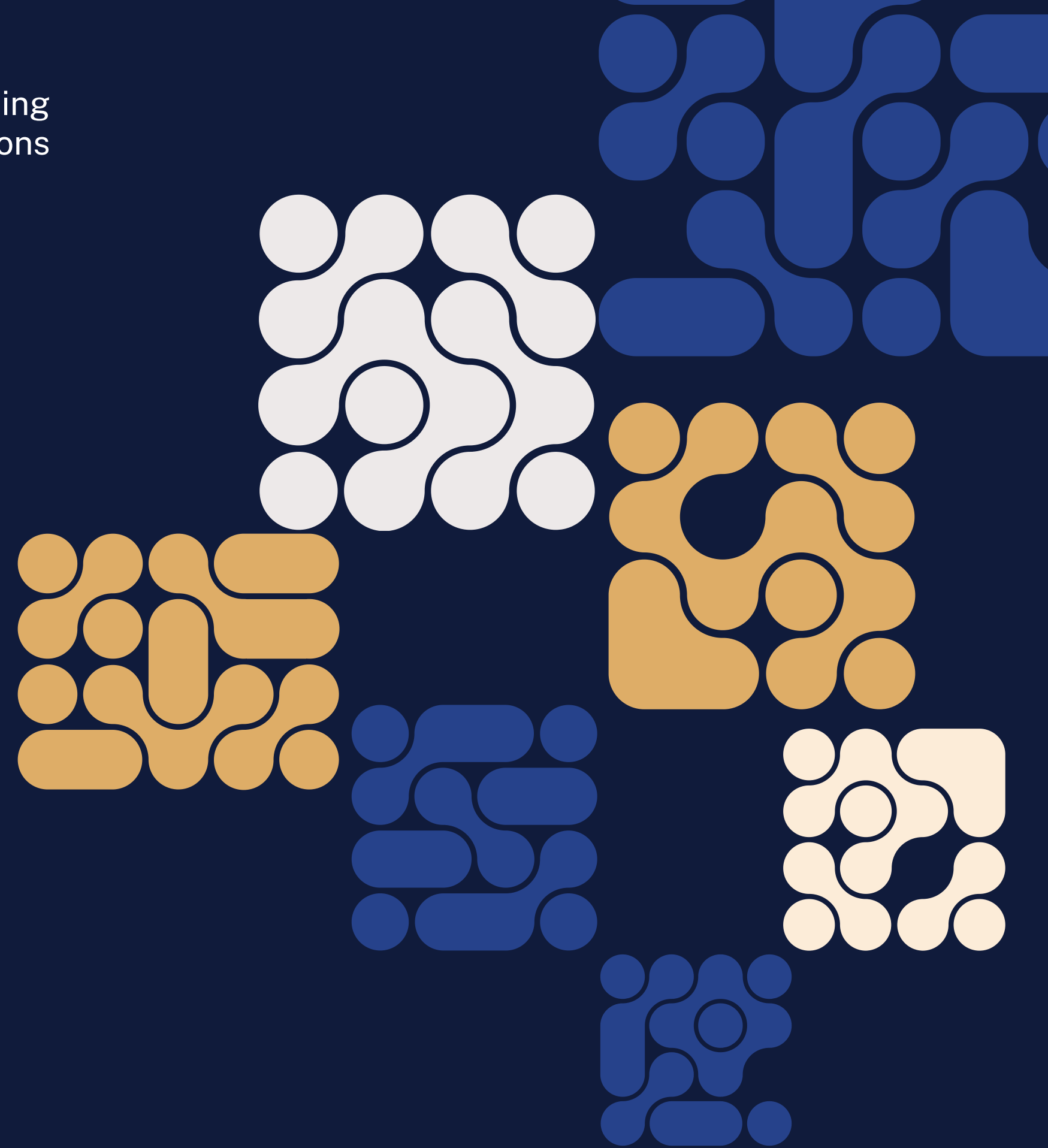
- LONG SHORT TERM MEMORY RECURRENT NEURAL NETWORKS (LSTM-RNN)
- INTRO TO HDLS
 - CHISEL
 - SYSTEM VERILOG

Presenter

Shahzaib Kashif

Co-Presenter

Shayan Hassan Baig



RECAP QUIZ TIME



LONG SHORT TERM MEMORY RECURRENT NEURAL NETWORKS (LSTM-RNN)

Quillbot Example

 **Grammar Checker** 

Apps and Extensions 

Upgrade to Premium  

Paraphraser

Grammar Checker

AI Detector

Plagiarism Checker

AI Humanizer

+




Untitled document

English (US) 

Enter writing task 

>>

Enter at least 25 words to see score.

1 All 

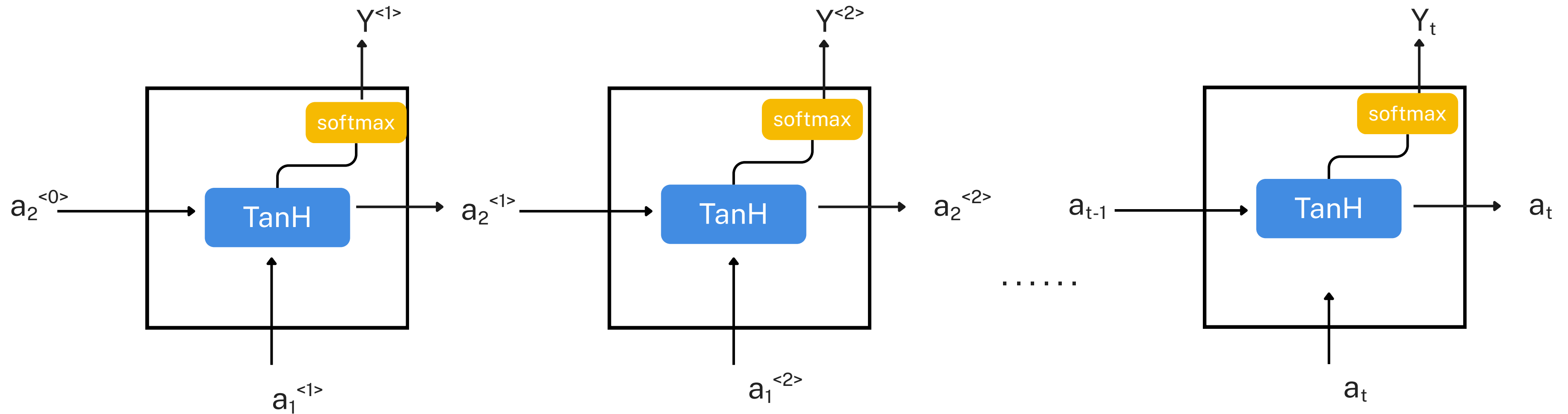
Change the pronoun 

~~her~~ **his** wife

Mr. Talha goes on a business trip every Wednesday, and one day during a summer break, ~~her~~ **his** wife was very angry.

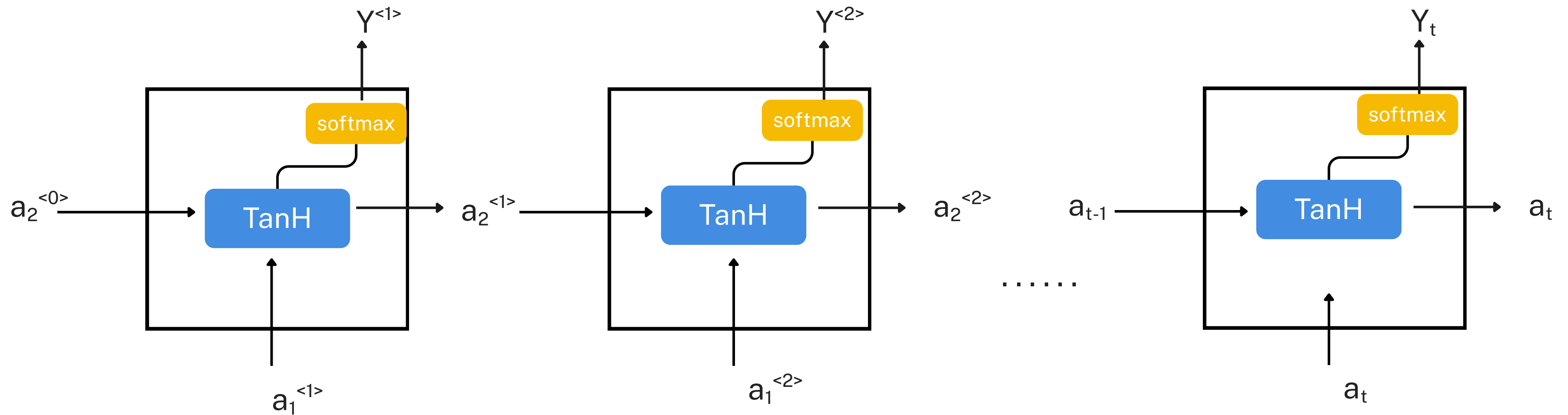
Accept Ignore

LSTM-RNN



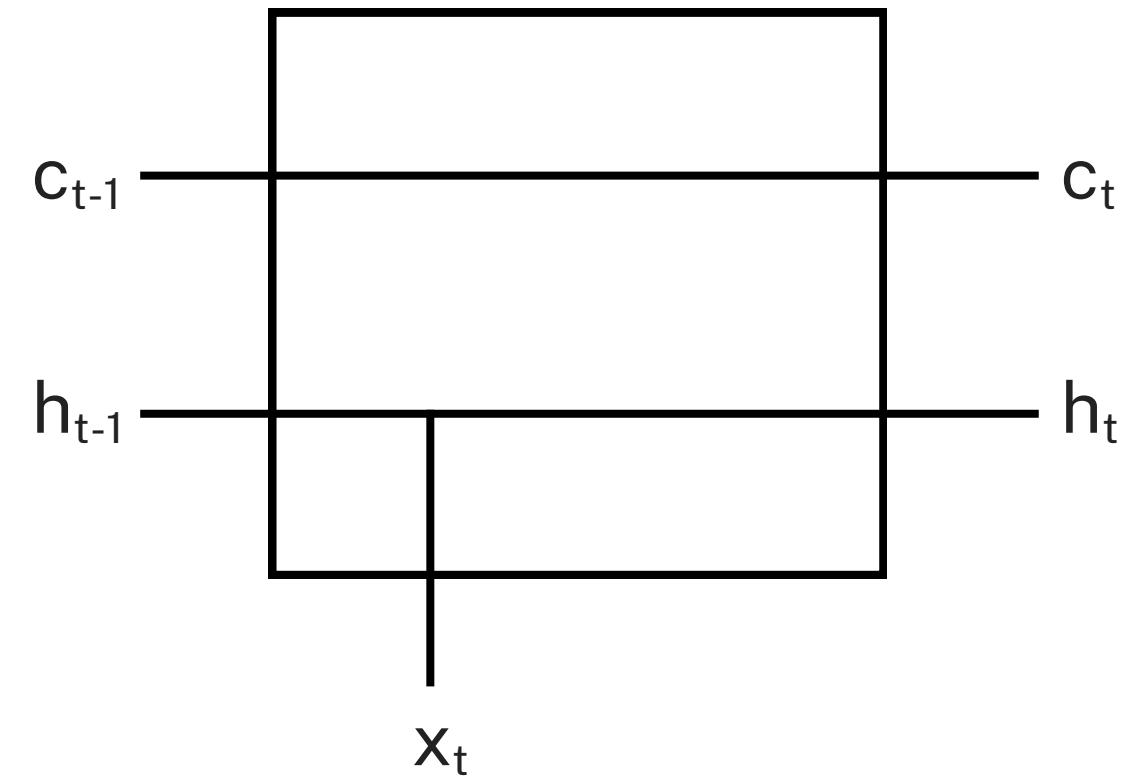
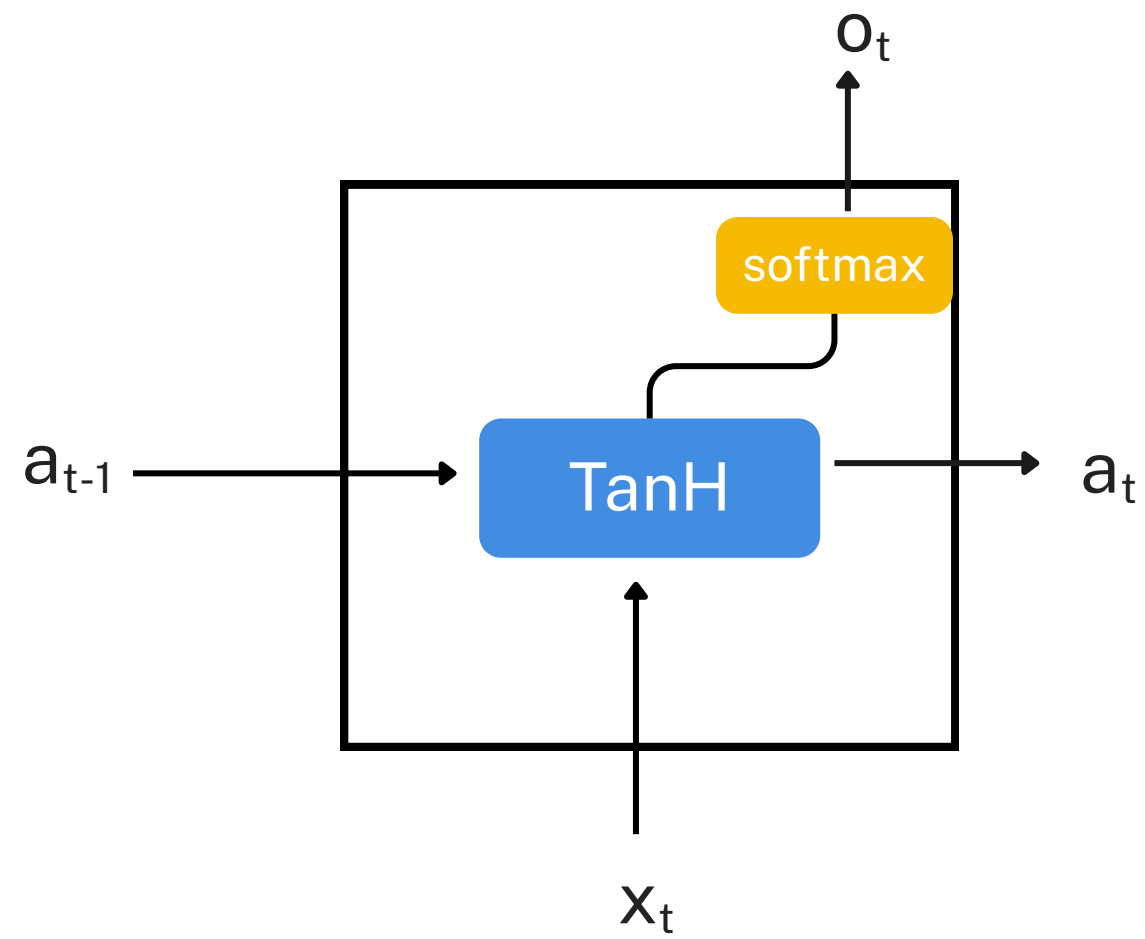
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LSTM-RNN

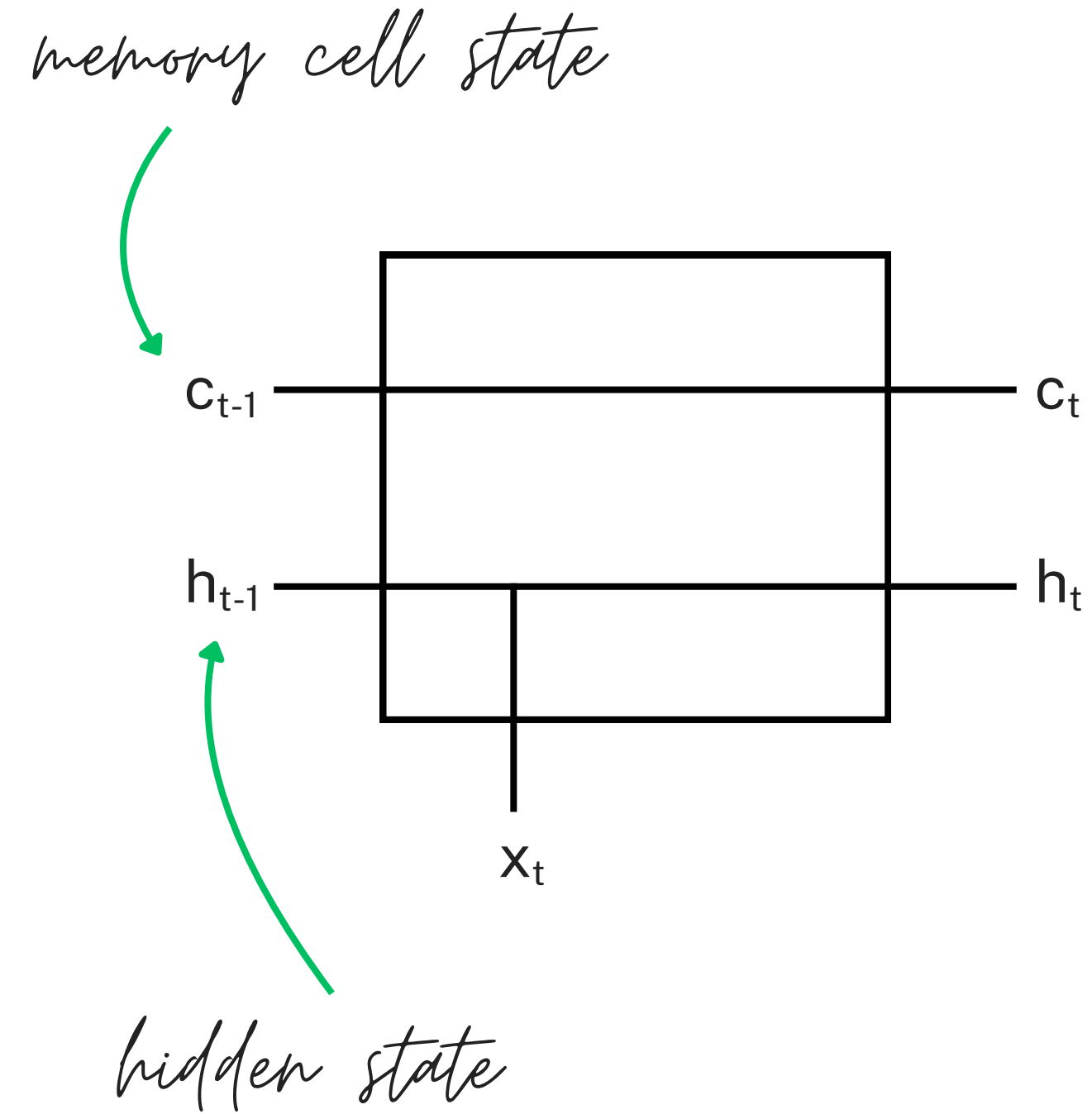
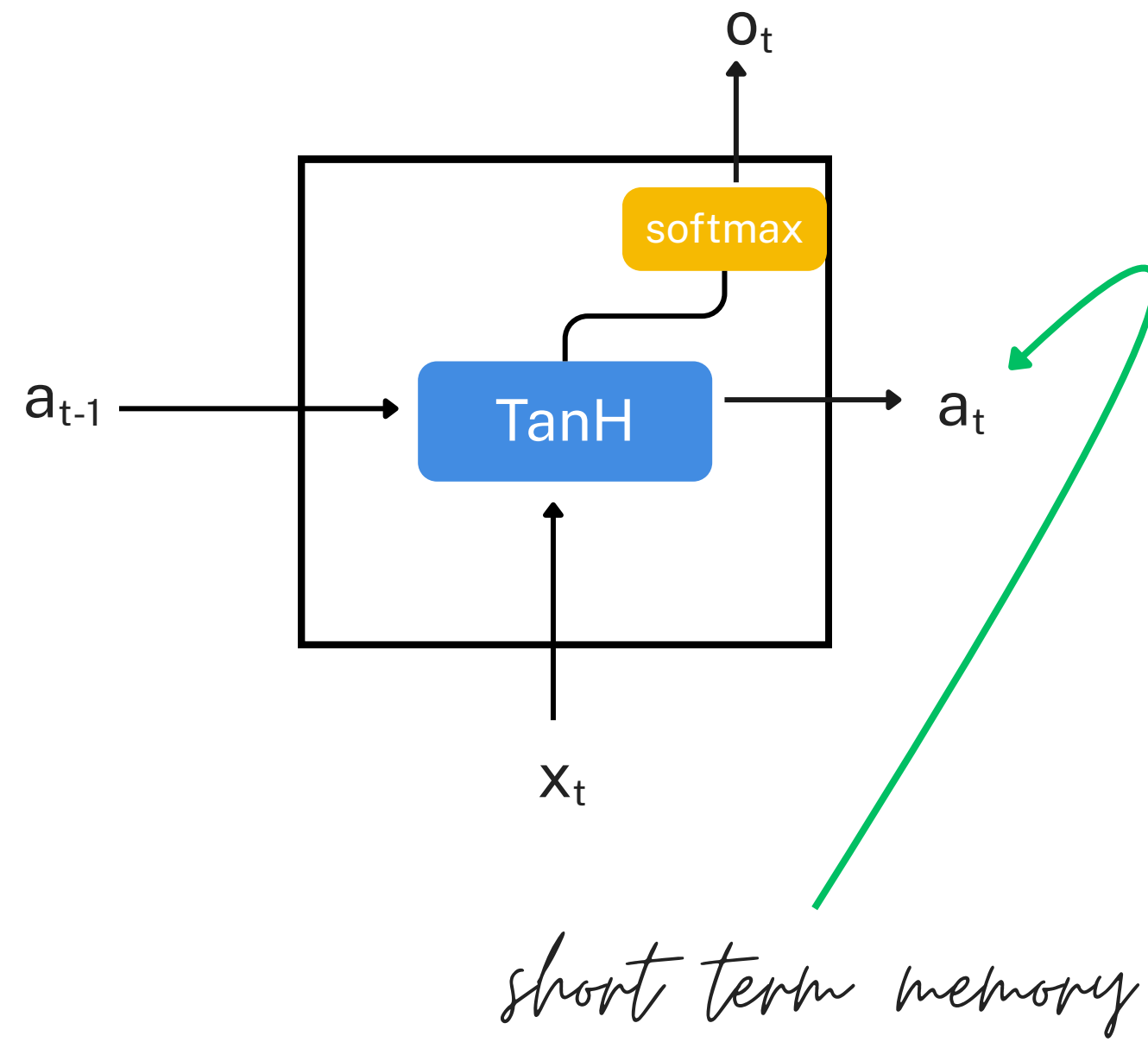


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LSTM-RNN

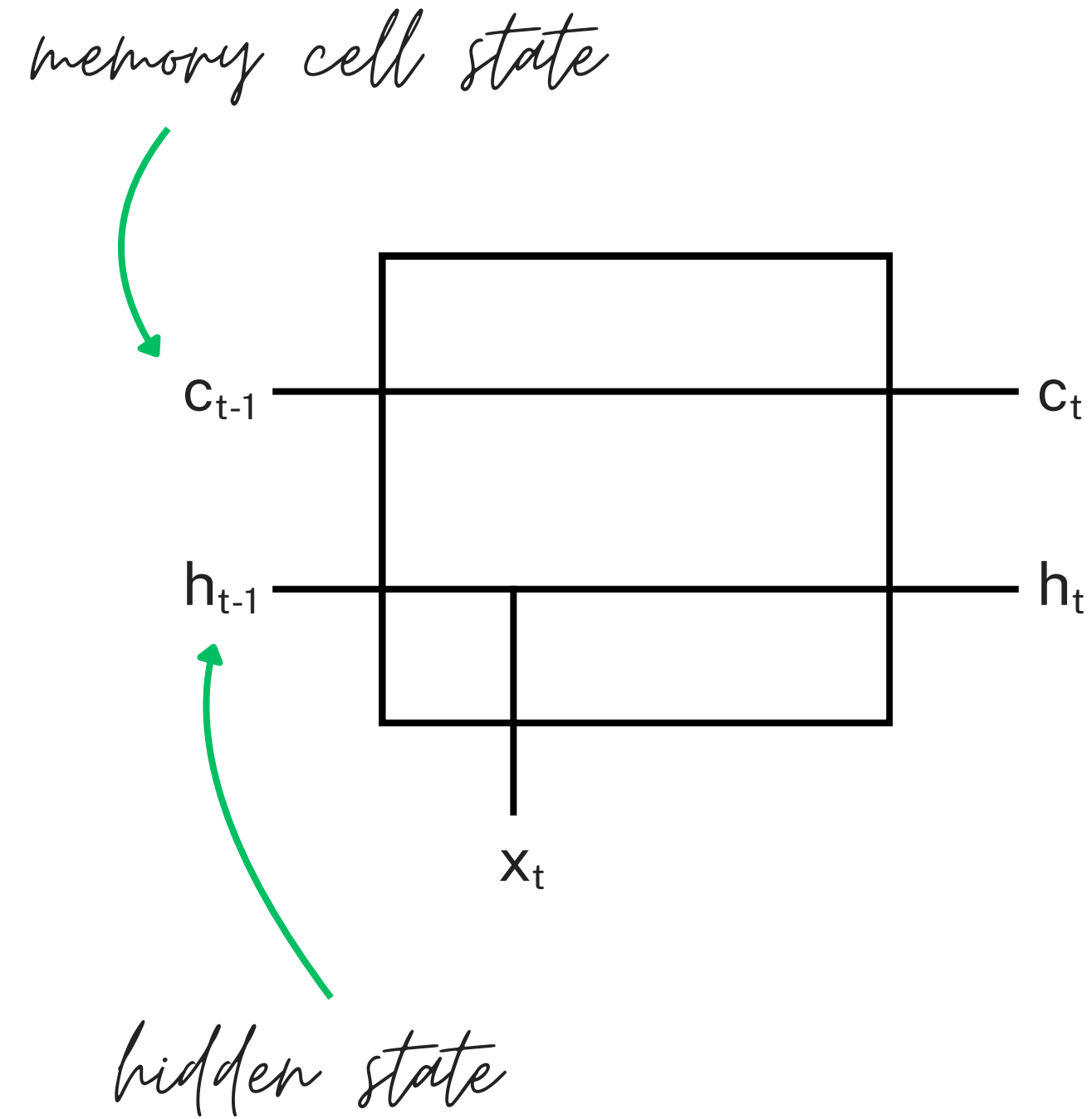
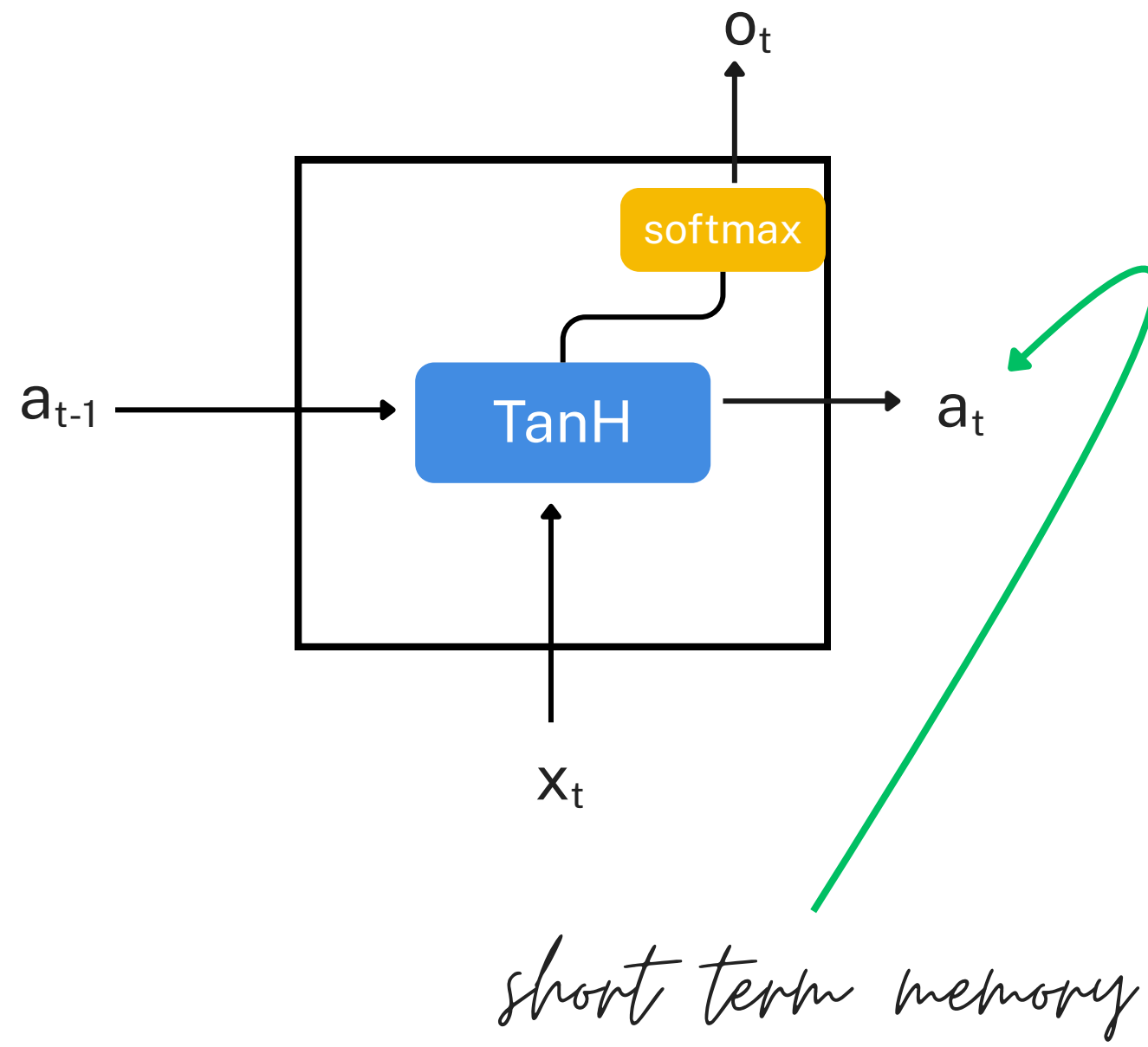


LSTM-RNN

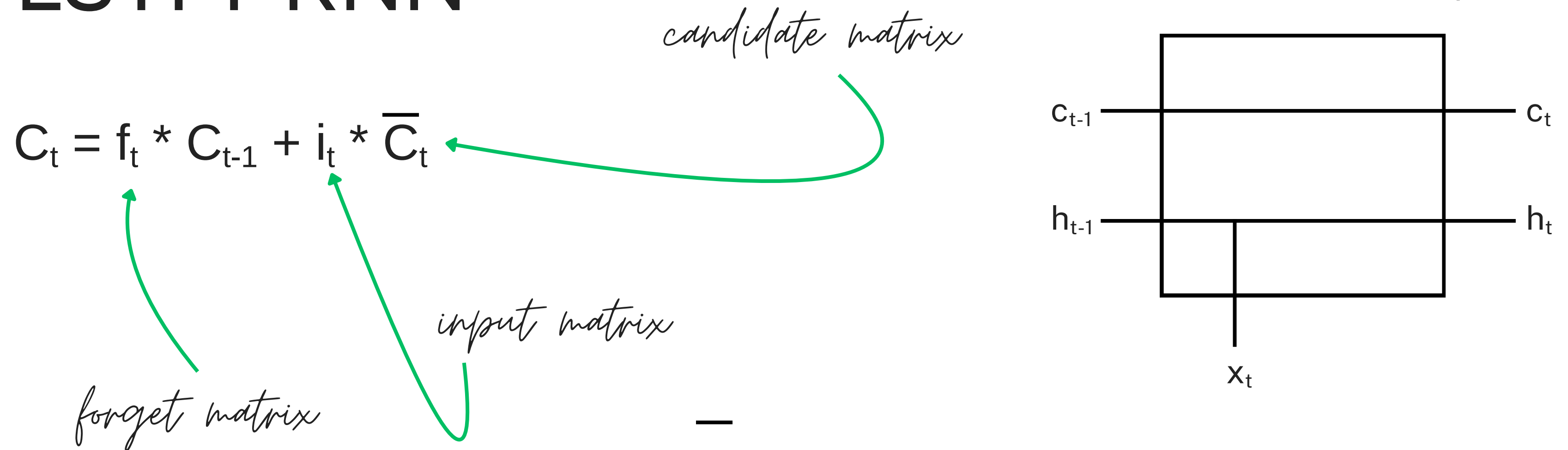


LSTM-RNN

Gates:
1 forget gate
2 input gate
3 output gate



LSTM-RNN



Mr. Talha goes on a business trip every Wednesday, and one day during summer break, **her** wife was very angry.

Ms Hafsa is Mr. Talha's friend, and she is a businesswoman **himself**.

LSTM-RNN

$$C_t = f_t * C_{t-1} + i_t * \bar{C}_t$$

forget matrix

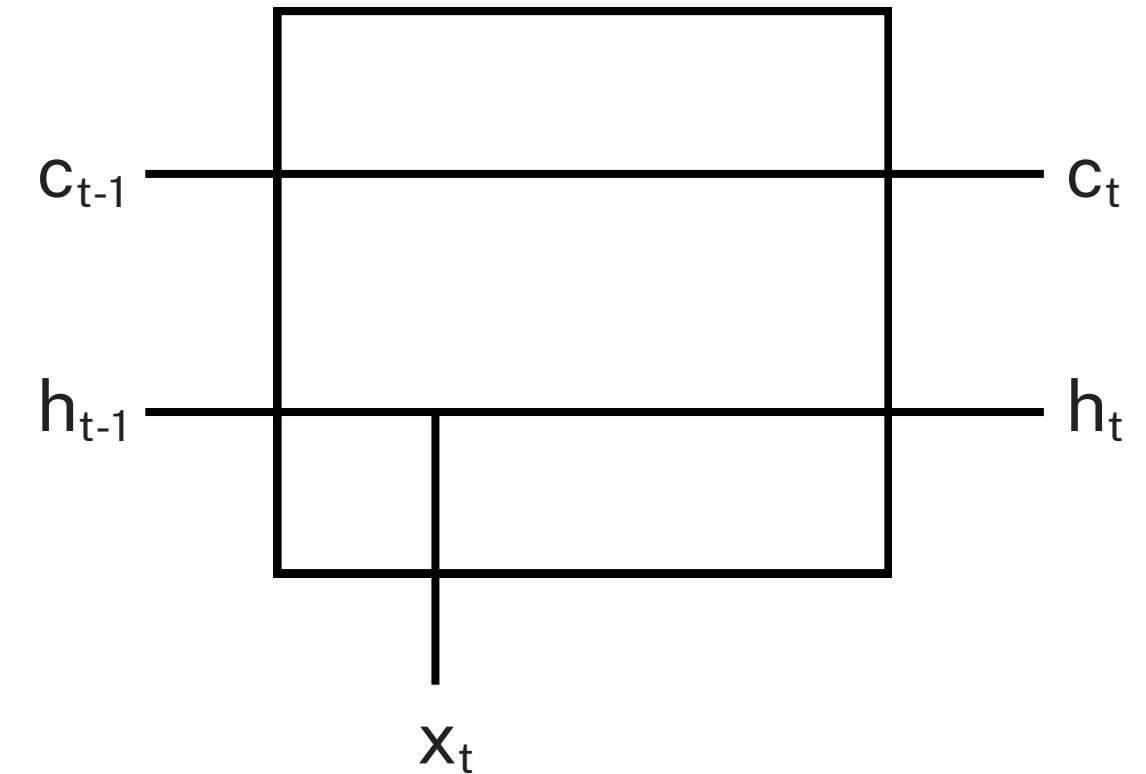
candidate matrix

input matrix

$$\bar{C}_t = \text{tanh}(W_c * h_{t-1} + W_c * x_t + b_c)$$

0 → discard

1 → allow to pass



LSTM-RNN

Mr. Talha goes on a business trip every Wednesday, and one day during summer break,
her wife was very angry.

And tomorrow, he was need to go on a trip again.

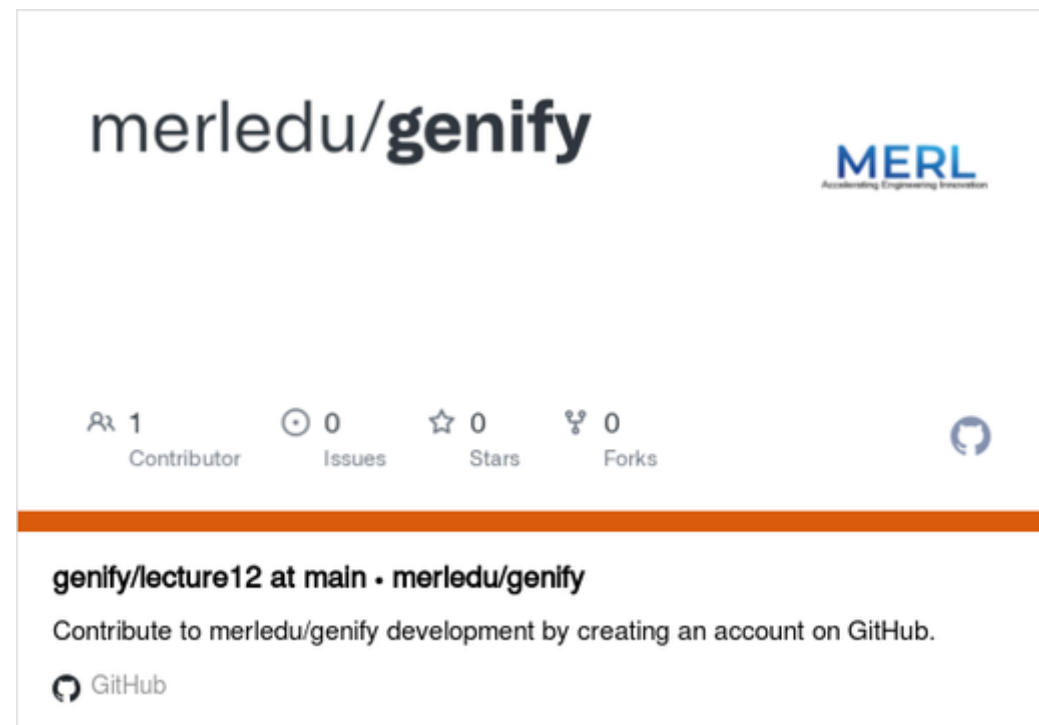
LSTM-RNN

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Python Walkthrough

<https://github.com/merledu/genify/tree/main/lecture12>



Reading Assignment

- The Ultimate Guide to Recurrent Neural Networks in Python
[20 min read]
- Applied LSTM: Use Cases, Types, and Challenges *[20 min read]*
- LSTM in Deep Learning: Architecture, Algorithm, And Applications *[25 min read]*



INTRODUCTION TO HARDWARE DESCRIPTION LANGUAGES

What are HDLs?

- **Definition:** HDLs are programming languages used to describe the structure, behavior, and timing of digital electronic circuits.
- **Key Purposes:**
 - Model hardware at various levels: Behavioral (high-level algorithms), RTL (Register-Transfer Level for synthesis), Gate-level (low-level primitives).
 - Enable simulation (test behavior virtually) and synthesis (convert to actual hardware like FPGAs or ASICs).
- **Why Not Software Languages?** HDLs handle concurrency (parallel operations in hardware) and timing, unlike sequential software langs like C/Python.
- **Examples of HDLs:** Verilog, VHDL, SystemVerilog, Chisel, High-Level Synthesis (HLS) tools like Vivado HLS.

What use HDLs in Digital Design?

- **Advantages:**

- *Abstraction:* Describe complex systems without drawing schematics manually.
- *Verification:* Simulate and debug before building hardware—saves time and cost.
- *Reusability:* Parametrized modules for scalable designs (e.g., a counter that works for 8-bit or 32-bit).
- *Automation:* Tools like simulators (ModelSim) and synthesizers (Vivado, Quartus) turn HDL into netlists or bitstreams.

- **Challenges:** Steep learning curve for concurrency/timing; potential for bugs if not verified properly.

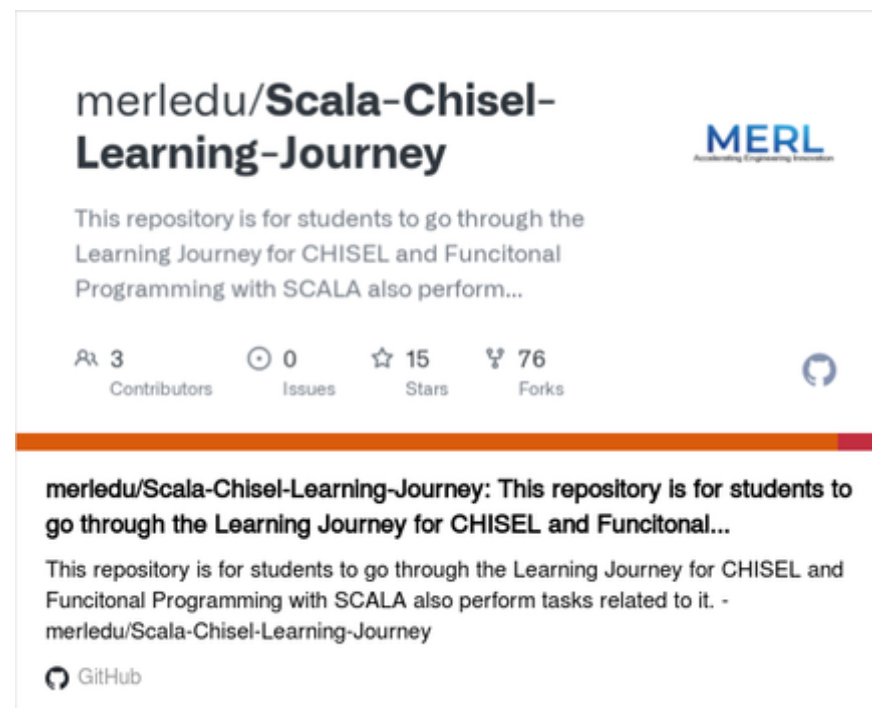
- **Real-World Use:** In CPUs (e.g., RISC-V), GPUs, IoT devices, and verification of chips at companies like Intel/AMD.

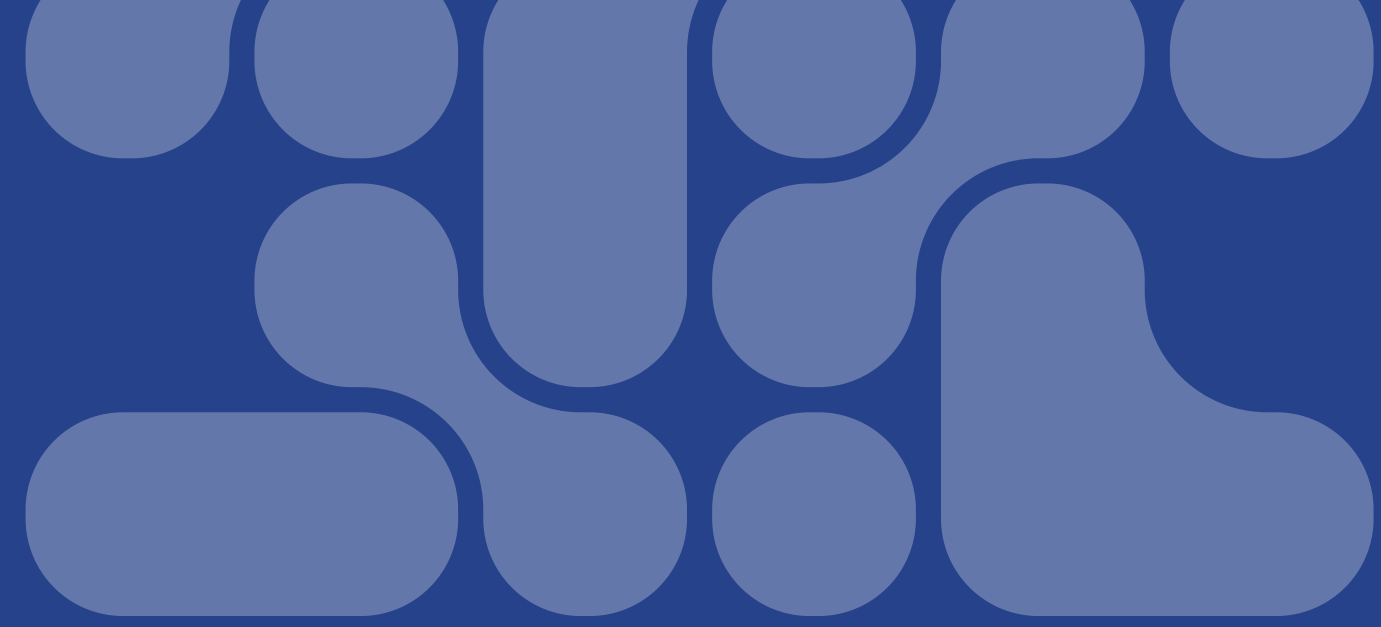
Basic Differences between CHISEL and System Verilog

Aspect	Chisel	System Verilog
Base Language	Embedded DSL in Scala (JVM-based, combines OOP and functional programming)	Standalone HDL, extends Verilog (C-like syntax)
Abstraction Level	Higher-level: Uses Scala features like classes, generics, and loops for concise, reusable code	Lower-level: Direct hardware constructs (wires, regs, always blocks) for fine-grained control
Productivity	More modern and productive; generates Verilog via FIRRTL intermediate representation	Verbose but industry-standard; great for verification with assertions and OOP extensions
Strengths	Parametrized designs, meta-programming (e.g., generate hardware dynamically); popular in research (e.g., RISC-V Boom)	Wide tool support, simulation/verification; used in production ASICs/FPGAs
Compilation	Scala program runs to output hardware description (Verilog)	Directly compiled/simulated/synthesized
Learning Curve	Steeper if new to Scala, but powerful for complex designs	Easier start if familiar with C, but can be tedious for large systems

Lab Walkthrough

<https://github.com/merledu/Scala-Chisel-Learning-Journey>





**Thank you for
your attention!**

Open floor for discussion.

